

# Functional Data Analysis on Monthly Sales

## Abstract

This study applies Functional Data Analysis (FDA) to explore monthly sales trends across U.S. states, aiming to uncover dominant temporal patterns and regional differences. Sales time series are smoothed using B-spline basis expansion, followed by functional principal component analysis (FPCA) to extract key dynamic features. K-means clustering on FPCA scores is then used to group states with similar sales behaviors. The results highlight the effectiveness of FDA in analyzing complex spatiotemporal sales data and provide insights into regional consumption dynamics.

## 1. Introduction

Understanding how consumer behavior evolves over time and across regions is critical for businesses, policymakers, and economists. In the U.S., substantial differences in seasonal demand, demographic structures, and economic conditions often lead to diverse sales patterns across states. Analyzing these temporal trends can inform inventory planning, targeted marketing, and resource allocation. However, traditional time series approaches often fall short when dealing with high-dimensional data that exhibit smooth and continuous temporal variation across multiple spatial units.

Functional Data Analysis (FDA) provides a flexible framework for modeling such data by treating each time series as a smooth function rather than a collection of discrete observations. This functional perspective enables the extraction of underlying temporal structures and facilitates comparisons across units. In particular, Functional Principal Component Analysis (FPCA) is widely used to reduce dimensionality while preserving essential features of the data, and clustering based on functional representations can reveal meaningful groupings in large-scale time series datasets.

This study leverages FDA to analyze monthly sales data across all U.S. states. The goals are threefold: to identify the main modes of variation in temporal sales patterns, to assess the effectiveness of FPCA in capturing these dynamics, and to explore the feasibility of clustering states based on their functional characteristics. By combining smoothing

techniques, FPCA, and unsupervised learning, this work offers a data-driven approach to uncovering structure in spatiotemporal consumer data.

## 2. Data Description

The analysis is based on the Superstore Sales dataset, publicly available on Kaggle (<https://www.kaggle.com/datasets/rohitsahoo/sales-forecasting/data>). The dataset contains detailed transactional records of 9,800 orders placed between January 2015 and December 2018. Key variables include Order Date (date of transaction), Sales (sales amount in USD), Region (one of Central, East, South, or West), and State (the U.S. state where the transaction occurred). A screenshot of the original dataset is as follows:

| Row ID | Order ID  | Order Date | Ship Date  | Ship Mode      | Customer                      | Customer Segment | Country       | City            | State      | Postal Code | Region | Product ID | Product Name     | Sales    |
|--------|-----------|------------|------------|----------------|-------------------------------|------------------|---------------|-----------------|------------|-------------|--------|------------|------------------|----------|
| 1      | CA-2017-1 | 16/06/2017 | 18/06/2017 | Standard Class | CG-12520 Claire Gutierrez     | Consumer         | United States | Henderson       | Kentucky   | 42420       | South  | FUR-BO-1   | Bush Somerset    | 261.96   |
| 2      | CA-2017-1 | 16/06/2017 | 18/06/2017 | Standard Class | CG-12520 Claire Gutierrez     | Consumer         | United States | Henderson       | Kentucky   | 42420       | South  | FUR-CH-1   | Honolulu Deluxe  | 731.94   |
| 3      | CA-2017-1 | 16/06/2017 | 18/06/2017 | Standard Class | DV-13045 Darrin Van Der Horst | Corporate        | United States | Los Angeles     | California | 90036       | West   | OFF-LA-1   | Self-Adhesive    | 14.62    |
| 4      | US-2016-1 | 18/10/2016 | 18/10/2016 | Standard Class | SO-20335 Sean O'Donnell       | Consumer         | United States | Fort Lauderdale | Florida    | 33311       | South  | FUR-TA-1   | Bretford Classic | 957.5775 |
| 5      | US-2016-1 | 18/10/2016 | 18/10/2016 | Standard Class | SO-20335 Sean O'Donnell       | Consumer         | United States | Fort Lauderdale | Florida    | 33311       | South  | OFF-ST-10  | Eldon Fold       | 22.368   |
| 6      | CA-2015-1 | 9/6/2015   | 14/06/2015 | Standard Class | BH-11710 Brosina Hc           | Consumer         | United States | Los Angeles     | California | 90032       | West   | FUR-FU-1   | Eldon Express    | 48.86    |
| 7      | CA-2015-1 | 9/6/2015   | 14/06/2015 | Standard Class | BH-11710 Brosina Hc           | Consumer         | United States | Los Angeles     | California | 90032       | West   | OFF-AR-1   | Newell 32        | 7.28     |
| 8      | CA-2015-1 | 9/6/2015   | 14/06/2015 | Standard Class | BH-11710 Brosina Hc           | Consumer         | United States | Los Angeles     | California | 90032       | West   | TEC-PH-1   | Mitel 5320       | 907.152  |
| 9      | CA-2015-1 | 9/6/2015   | 14/06/2015 | Standard Class | BH-11710 Brosina Hc           | Consumer         | United States | Los Angeles     | California | 90032       | West   | OFF-BI-10  | DXL Angle        | 18.504   |
| 10     | CA-2015-1 | 9/6/2015   | 14/06/2015 | Standard Class | BH-11710 Brosina Hc           | Consumer         | United States | Los Angeles     | California | 90032       | West   | OFF-AP-1   | Belkin F5C       | 114.9    |

Figure 1: screenshot of the original dataset.

To enable functional data analysis, monthly sales were aggregated at both the state and regional levels. States with fewer than five months of nonzero sales were excluded to ensure temporal consistency and data reliability. Two aggregated datasets were constructed accordingly: one by state and the other by region. A continuous time variable was created by converting year and month into decimal form (e.g., January 2015 as 2015.000; August 2016 as 2016.667), allowing for functional smoothing over time. Screenshots of the two aggregated datasets are provided below:

|    | Region  | Year | Month | Monthly_Sales | Time     |
|----|---------|------|-------|---------------|----------|
| 1  | Central | 2015 | 1     | 1533.966      | 2015.000 |
| 2  | Central | 2015 | 2     | 1233.174      | 2015.083 |
| 3  | Central | 2015 | 3     | 5827.602      | 2015.167 |
| 4  | Central | 2015 | 4     | 3712.340      | 2015.250 |
| 5  | Central | 2015 | 5     | 4044.522      | 2015.333 |
| 6  | Central | 2015 | 6     | 9374.107      | 2015.417 |
| 7  | Central | 2015 | 7     | 6740.574      | 2015.500 |
| 8  | Central | 2015 | 8     | 3022.183      | 2015.583 |
| 9  | Central | 2015 | 9     | 34254.866     | 2015.667 |
| 10 | Central | 2015 | 10    | 8965.757      | 2015.750 |

|    | State   | Year | Month | Monthly_Sales | Time     |
|----|---------|------|-------|---------------|----------|
| 1  | Alabama | 2015 | 4     | 1224.880      | 2015.250 |
| 2  | Alabama | 2015 | 5     | 180.930       | 2015.333 |
| 3  | Alabama | 2015 | 10    | 2216.800      | 2015.750 |
| 4  | Alabama | 2015 | 11    | 14.670        | 2015.833 |
| 5  | Alabama | 2015 | 12    | 2501.810      | 2015.917 |
| 6  | Alabama | 2016 | 3     | 518.490       | 2016.167 |
| 7  | Alabama | 2016 | 6     | 976.290       | 2016.417 |
| 8  | Alabama | 2016 | 7     | 457.040       | 2016.500 |
| 9  | Alabama | 2016 | 9     | 1940.150      | 2016.667 |
| 10 | Alabama | 2017 | 5     | 1007.680      | 2017.333 |

Figure 2: screenshots of the two aggregated datasets.

### 3. Functional Representation of Temporal Sales Data

#### 3.1 Regional Sales Trends and Functional Smoothing

Monthly aggregated sales by region from 2015 to 2018 were first visualized using a scatter plot to reveal overall temporal variation. Figure 3 shows raw monthly sales for the four U.S. regions (Central, East, South, and West), each color-coded for distinction. Visual inspection suggests potential upward trends in some regions and seasonal spikes around the end of each year.

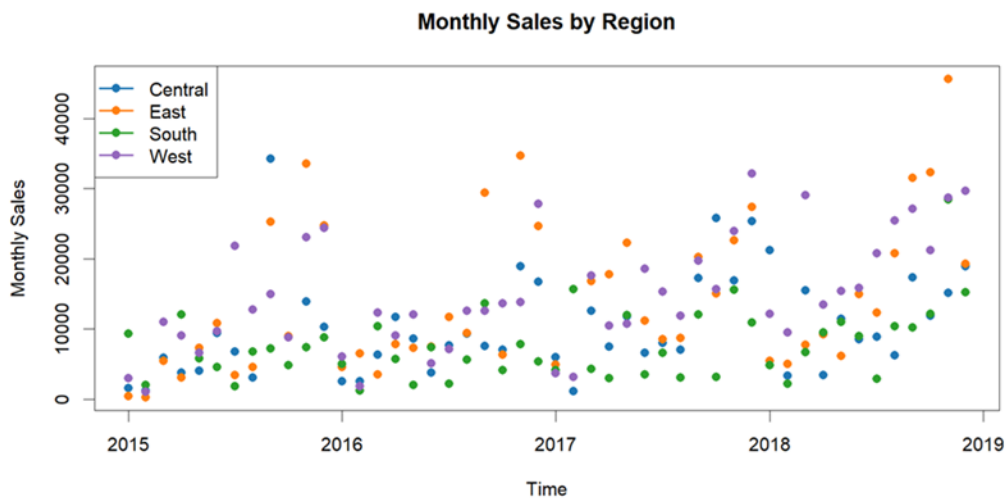


Figure 3: scatter plot of monthly sales by regions.

To extract underlying sales dynamics and remove short-term fluctuations, functional data analysis techniques were applied. Specifically, regional monthly sales series were transformed into smooth functional curves using B-spline basis expansion. Each sales curve was expressed as a linear combination of 14 cubic B-spline basis functions, selected via cross-validation. A regularization parameter  $\lambda = 0.8$  was employed to balance smoothness and fidelity to the data.

Figure 4 presents the resulting smoothed sales curves superimposed on the scatter plot. The smoothed representations clearly highlight region-specific patterns. The East and West regions exhibit a strong and consistent upward trend, particularly after 2017, indicating expanding consumer activity. The Central region shows moderate growth with some seasonality, while the South remains relatively flat throughout the period, with consistently

lower sales levels. These smooth curves serve as compact summaries of temporal dynamics across regions.

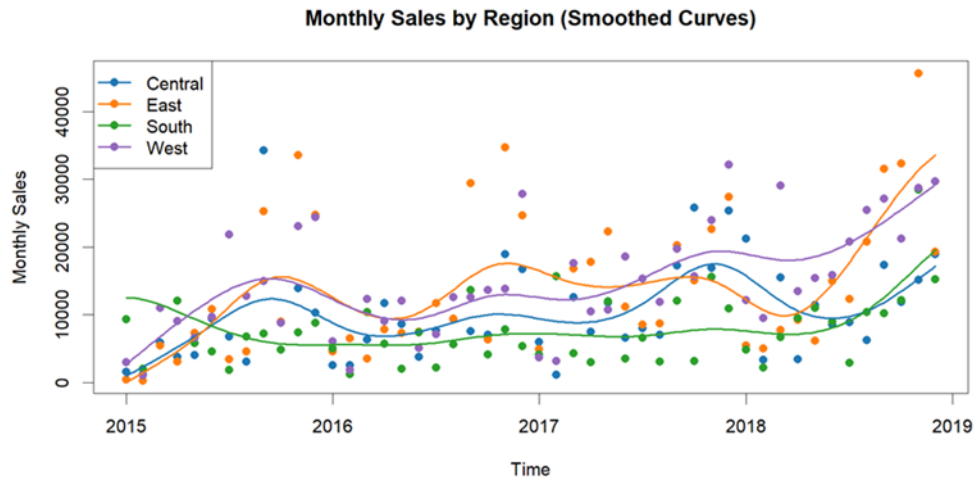


Figure 4: smoothed curves of monthly sales by regions.

### 3.2 State-Level Sales Trends and Functional Smoothing

The scatter plot of monthly sales by state reveals substantial heterogeneity in sales trajectories across the country. Figure 5 shows the raw data, where patterns range from sharp growth in certain states to stagnation in others. Some degree of seasonality is visible in a few high-volume states, but noise and irregularity dominate many of the series.

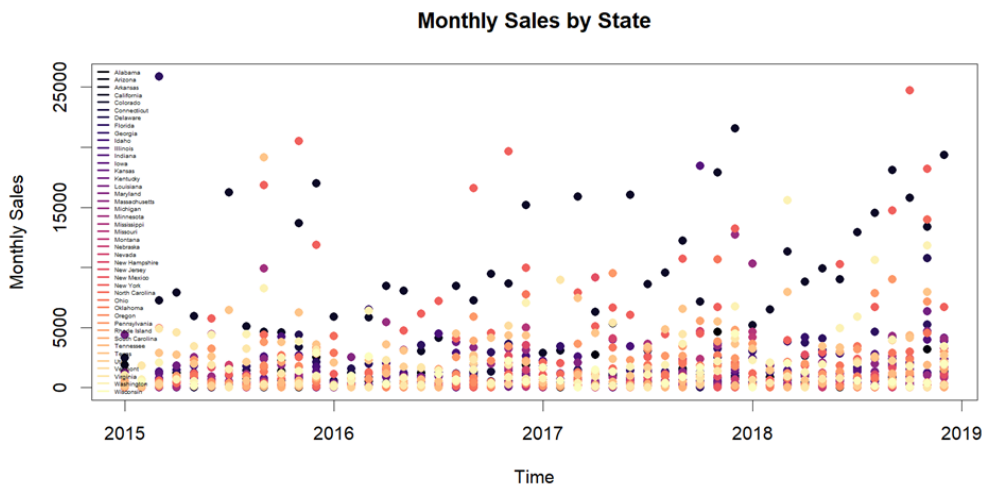


Figure 5: scatter plot of monthly sales by states.

To better capture these differences, the same functional smoothing procedure was applied at the state level. Only states with at least five months of nonzero sales were included in the analysis to ensure meaningful curve estimation. Figure 6 displays the smoothed functional sales curves for the retained states. These curves reveal clear distinctions in temporal dynamics. High-population states such as California and New York exhibit well-defined upward trends and seasonal cycles, suggesting robust market performance. In contrast, most other states show flat or mildly increasing sales paths, with limited structural variation. These smooth representations reduce noise and highlight latent similarities and differences across states, forming the basis for principal component analysis and clustering in the next step.

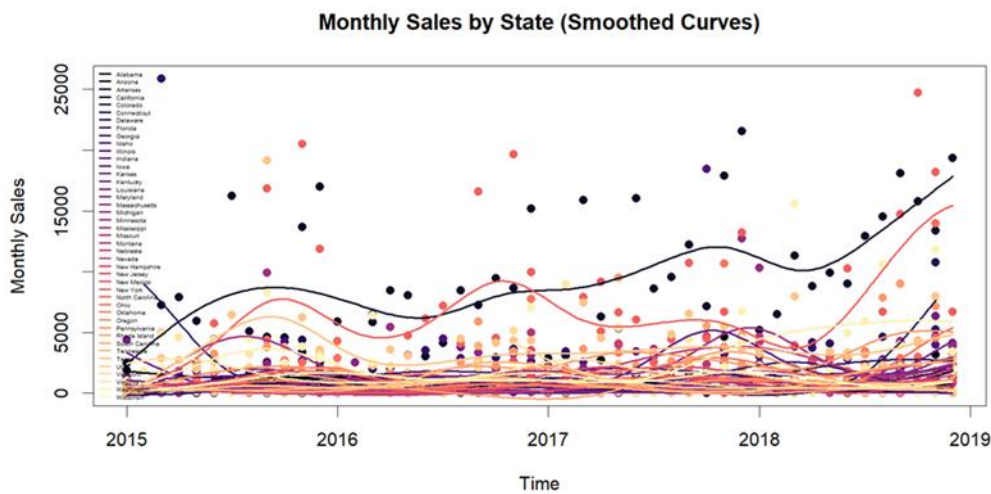


Figure 6: smoothed curves of monthly sales by states.

## 4. Functional Principal Component Analysis and Clustering

### 4.1 Functional Principal Component Analysis

To explore the dominant modes of variation among state-level sales trajectories, Functional Principal Component Analysis (fPCA) was applied to the smoothed functional curves obtained above. fPCA decomposes each function into a mean component and a linear combination of orthogonal eigenfunctions, effectively capturing the major directions of variability in the dataset. This method not only retains the temporal structure of the original time series but also enables significant dimensionality reduction.

The first four principal components were selected, as they together accounted for over 95% of the total variation across states. The first functional principal component (fPC1) exhibits a broadly positive shape with two large humps around months 10 and 30, corresponding approximately to late 2015 and mid-2017. This component captures the overall average sales trend across states. The second component (fPC2) presents a wave-like oscillating pattern, reflecting seasonal or cyclical fluctuations in monthly sales. States with strong seasonal sales behavior load heavily on this component. The third and fourth components describe idiosyncratic short-term effects, though they contribute less to total variation.

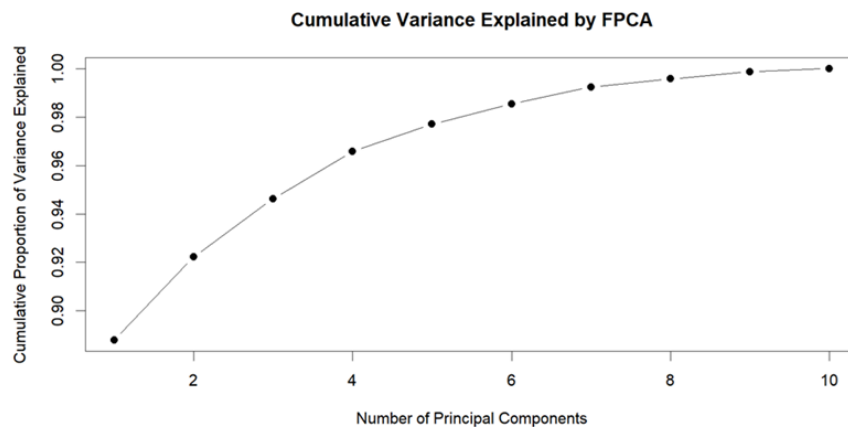


Figure 7: cumulative variance explained by fPCA.

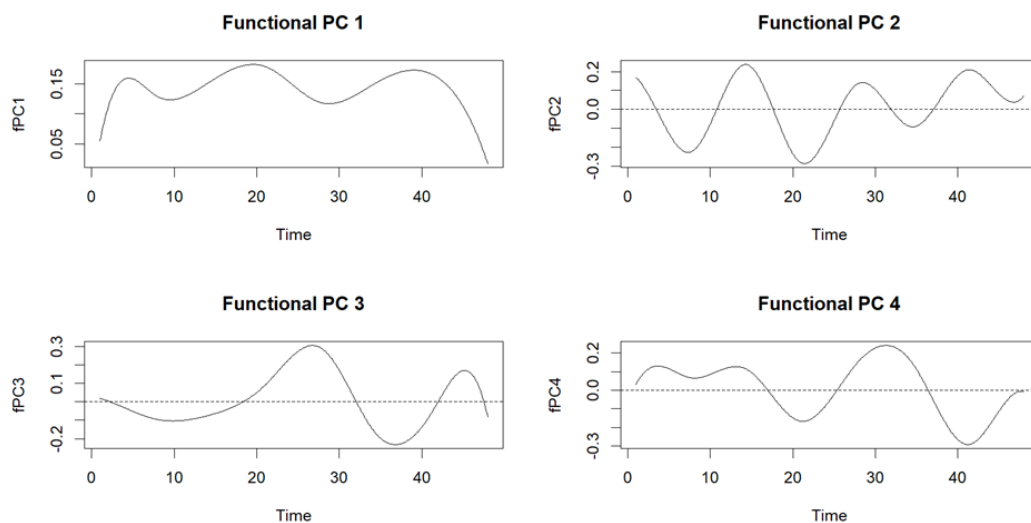


Figure 8: first four principal components.

## 4.2 Clustering States

To further interpret the variation captured by fPCA, k-means clustering was applied to the four-dimensional fPCA scores. This allowed for grouping states with similar temporal sales patterns based on their functional characteristics. The elbow method was used to determine the optimal number of clusters, with the result suggesting three distinct clusters.

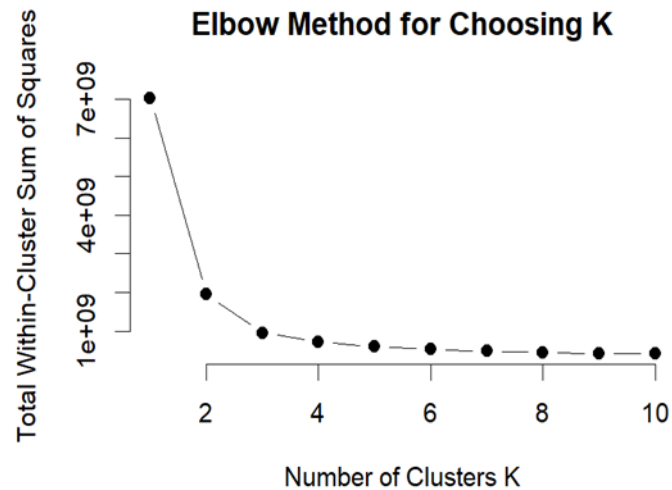


Figure 9: elbow method for choosing K.

Figure 10 shows the clustering results visualized in the PC1 and PC2 space, highlighting the separation between clusters. The clustering revealed meaningful segmentation of U.S. states. The first cluster comprises nine states, including Florida, Illinois, and Michigan. These states are characterized by moderate to high sales volumes and display visible seasonality along with steady upward trends. The second cluster contains only two states, California and New York. These states are distinct both in terms of their sales magnitude and temporal structure, exhibiting consistently strong growth and prominent seasonal cycles. The third cluster, which includes 38 states, encompasses the majority of the U.S. and is marked by relatively flat or mildly increasing sales trajectories with low seasonal variability. The clustering results underscore the effectiveness of fPCA in simplifying complex functional data while preserving meaningful differences for downstream analysis.

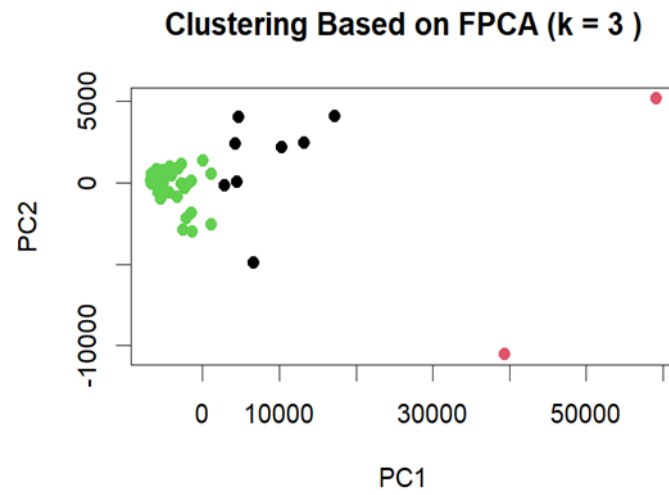


Figure 10: clustering results visualized in the PC1 and PC2 space.

| State                | Cluster |
|----------------------|---------|
| Florida              | 1       |
| Illinois             | 1       |
| Michigan             | 1       |
| Ohio                 | 1       |
| Pennsylvania         | 1       |
| Texas                | 1       |
| Virginia             | 1       |
| Washington           | 1       |
| California           | 2       |
| New York             | 2       |
| Alabama              | 3       |
| Arizona              | 3       |
| Arkansas             | 3       |
| Colorado             | 3       |
| Connecticut          | 3       |
| Delaware             | 3       |
| District of Columbia | 3       |
| Georgia              | 3       |

Figure 11: screenshot of clustering results.

## 5. Conclusion

This project applied Functional Data Analysis to extract meaningful temporal structures from U.S. monthly retail sales data between 2015 and 2018. By smoothing raw sales series using B-spline basis functions, long-term trends and seasonal patterns were revealed in both regional and state-level data. Functional Principal Component Analysis successfully captured dominant variation modes, allowing dimensionality reduction while retaining essential information. Subsequent clustering based on functional representations identified distinct groups of states, highlighting heterogeneity in consumer behavior across the country.

The functional modeling framework enabled clearer interpretation of complex, noisy sales trajectories. Unlike traditional time series methods, this approach emphasized structural patterns and allowed consistent comparison across geographic units. The identification of a small number of interpretable clusters also suggests potential applications in market segmentation, policy design, and regional strategy development.

However, several limitations remain. The analysis is restricted to two datasets without incorporating explanatory factors such as population size, economic indicators, or promotional activities. Additionally, the smoothing and clustering procedures involve choices, such as the number of basis functions and clusters, that may influence results. These constraints suggest that while the findings offer valuable insights into temporal sales variation, caution is warranted in generalizing conclusions beyond the given context.